

OLM Smoother v2 Manual



OLM, Inc. OLM Digital, Inc.
Research and Development

Introduction

OLM Smoother is an After Effects plug-in specialized in smoothing cel animation drawings. The v2 supports color space conversion and improves the smoothing of diagonal lines and so on.



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Original, OLM Smoother v1, OLM Smoother v2

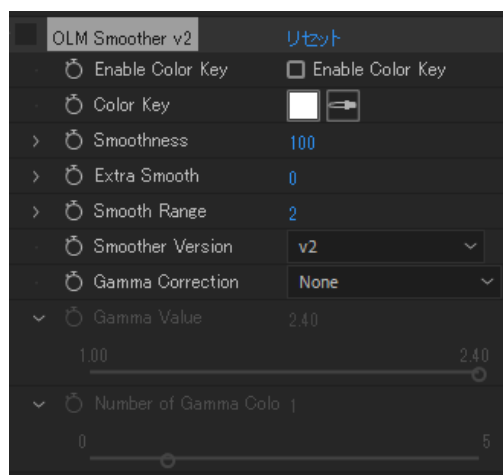
Release Notes

- v2
 - Add Smoothness
 - Add Extra Smooth
 - Add Smoother Version
 - Add Gamma Correction
 - Add Gamma Value
 - Add Number of Gamma
 - Add Gamma Color
 - MFR support
 - 32 bit support

Usage

To apply the effect, use the menu: Effects → OLM Plug-ins → OLM Smoother.

Apply it to a cel drawing to smooth the lines while using the color key parameters to get a transparent background. In case there are gradations in the colors you can use the Smooth Range parameter. You can use Gamma Correction to adjust line thickness.



OLM Smoother parameters

Use Color Key

When “Use Color Key” parameter is on, you can choose a color to key with the “Color Key” parameter before applying OLM Smoother.



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Before/After applying color keying

Smoothness and Extra Smooth

Smoothness is the strength of the smoothing. When 0, nothing gets smoothed. Default is 100, maximum smoothing is applied. Extra Smooth parameter adds smoothing to get even smoother lines.

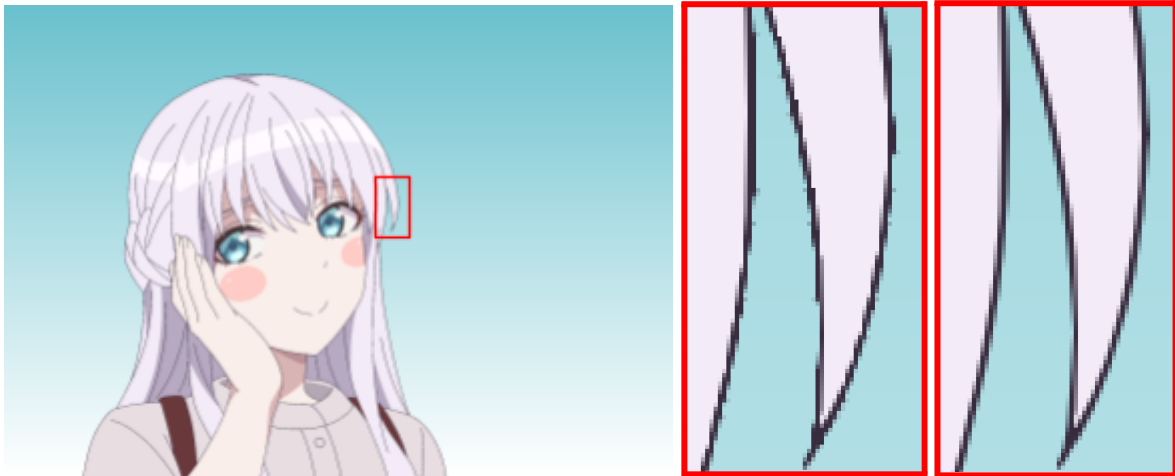


Smooter Version

You can choose a Smooter Version value: v1 or v2. This only affects how the [Gamma Correction](#) is applied. When using v1, we do not linearize the color prior to processing.

Smooth Range

When the layer you want to apply OLM Smooter contains gradations, the lines don't get as smooth as you expect. By tuning the "Smooth Range" parameter, you can get smoother lines in that case.



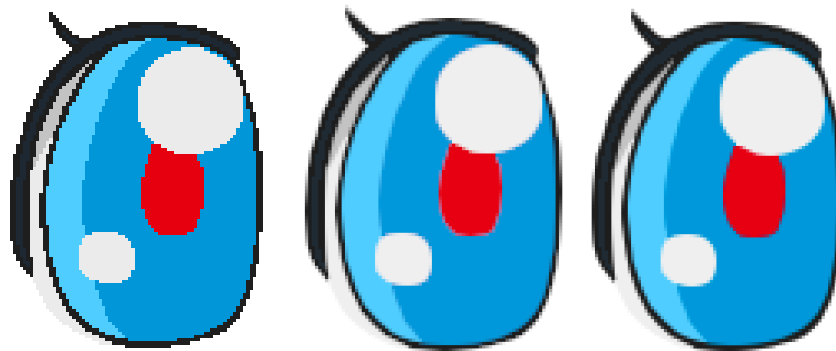
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Input、Do Smooth Range=0、Do Smooth Range=6

This parameter sets a range where we consider the color to be the same. The parameter range is [0,255]. When 255 the output is identical to the input. Usually the valid range that results in smooth lines is between 0 and 20.

Gamma Correction

OLM Smooter processing is assuming the input is sRGB. So we start by converting sRGB to linear, smooth the image before converting back to sRGB. The Gamma Correction parameters allow adding a gamma conversion after the conversion to linear conversion, before and after the smoothing.

The merit of sRGB conversion is to get a better blend quality of the colors and thinner the black lines. By default we use sRGB conversion because OLM artists like that result. We give a comparison of Gamma correction results below.



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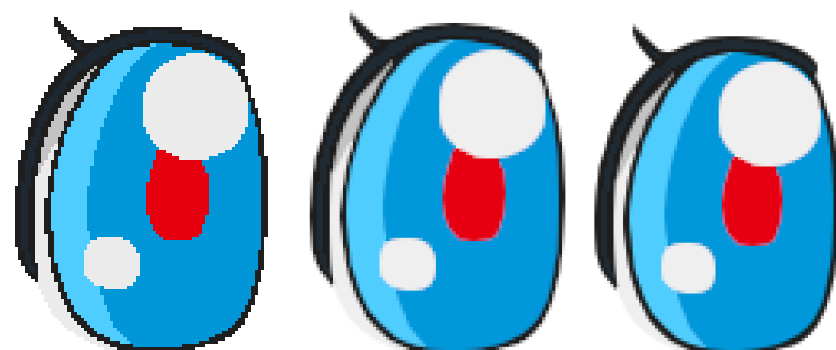
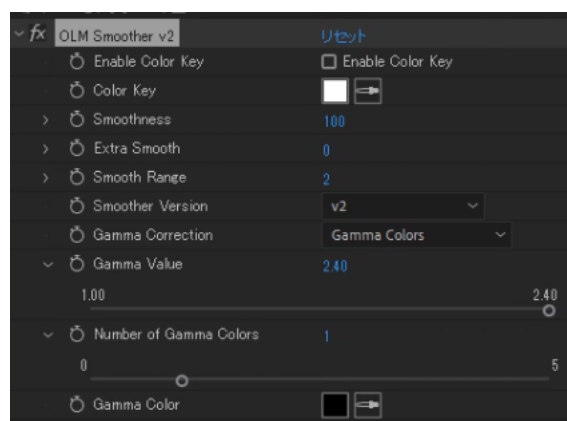
Input, No Gamma Correction, Gamma correction=AllColors + Gamma Value=2.4

You can see that in the middle image, the red and blue colors are mixing correctly, the lines are thinner than the lines on the right image. On the right image you can see some dark fringe around the red pupil.

If you don't want to use sRGB to Linear conversion, set the Smoother Version parameter to v1.

Gamma Correction may be set in All Colors mode where all colors have gamma correction applied. The bigger Gamma Value parameter is, the thicker lines and the more black fringe at extreme color blend like blue/red color blend.

If you want to keep black line thickness while have nice color blend, you can use the Gamma Colors mode of the Gamma Correction parameter. By adding only the black lines in the Gamma Colors list, you can obtain the following results.



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Smoother Version = v2

Gamma Correction = Gamma Colors, black only

Input, Gamma Value=1, Gamma Value = 2.4

Note

After Effects plugins from OLM Open Tools use Visual C++ 2019 runtime. They require the installation of Visual C++ Redistributable that supports Visual C++2019. After Effects usually ships with a compatible version, but if the plugin does not work please install the [latest Visual C++ Redistributable](#) yourself.